

THOMPSON RIVERS UNIVERSITY

Augmented Sample Size Calculation With Truncated Data

By

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Thesis Proposal

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1.Introduction

Parameter estimation is a type of inference that is a central topic in statistical research. Estimation can be made more challenging such as when observations are truncated or censored. This is further complicated as the sample size, an integer, may not seem as appropriate when an observation is presented only within one of these kinds of intervals. Building upon prior work in this area, this study proposes an iterative algorithm that augments the sample size calculation.

To illustrate the motivation for this framework, we revisit Munasinghe (2023) the treatment of the sample size in the simulated data sets $R = 500$ of the BVN model, which highlights the limitations of fixed-value adjustments.

Munasinghe (2023) employed the concept of sample size in her thesis. Within the dataset using BVN model, she defined the proportion of missing values and proposed the formula

$$n_{\text{missing}} = 0.5p \times n,$$

Missingness is introduced into BVN datasets considering a missing proportion p . n_{missing} denotes the number of missing observations for each variable. However, Munasinghe (2023) findings suggest that the estimation of both the intercept and slope lacks accuracy. Our findings suggest effective sample size should be adjusted using a fixed value that does not adequately reflect the truncated or censored variability. These limitations highlight the necessity of developing a more general framework for estimating sample size across diverse scenarios.

To situate this contribution within the broader scholarly context, it is essential to review prior research on truncation, censoring, and sample size estimation.

2.Literature Review

Estimating population size under truncation is a long-standing challenge in statistics. Censored observations only partial information is available for a specified left bound and right bound as infinity; truncation implies that all observations outside specified bounds are completely excluded from the sample (Fisher, 1941; Efron and Petrosian, 1999). Early work addressed this problem through likelihood-based approaches, including binomial and truncated multinomial formulations, extending classical models from life testing and capture–recapture studies (Sanathanan, 1972; Chao, 1987). These methods rely heavily on asymptotic theory to justify constrained likelihood estimation and emphasize the necessity of jointly accounting for uncertainty in both the population size N and the model parameter θ (Bickel and Doksum, 2001). Their unified treatment of truncated multinomial data, bridging discrete and continuous cases, provides important methodological insights. Although the present thesis does not adopt these frameworks directly, their asymptotic perspective strongly informs our approach to hypothesis and parameter estimation under truncation.

Related challenges arise in clinical trial settings, where censoring complicates sample size determination and inference (Lagakos, 1979). Clinical trials with time-to-event outcomes often face additional difficulties due to staggered enrollment, censoring, and uncertain event rates. Shen and Cai (2003) proposed an adaptive design that enables sequential sample size re-estimation based on weighted linear rank statistics, allowing trials to maintain desired power while controlling the overall type I error rate. Their framework explicitly addresses the misalignment between internal subject time and external monitoring time, which undermines naive re-estimation strategies. While we do not adopt their exact adaptive framework, their asymptotic theory provides information for developing robust parameter es-

timization procedures under censoring (Jennison and Turnbull, 2000). In contrast, our work employs an EM-based approach to handle incomplete observations.

Parallel concerns emerge in the modeling of extreme events, where truncation and censoring due to measurement limits pose substantial difficulties for parameter estimation in heavy-tailed distributions (Coles, 2001). Modeling extreme events is essential in domains such as finance, hydrology, and reliability engineering, where tail behavior drives risk assessment (Embrechts, Klüppelberg, and Mikosch, 1997). The Generalized Pareto Distribution (GPD) is a standard model for threshold exceedances within the Peaks Over Threshold (POT) framework (Pickands, 1975; Davison and R. L. Smith, 1990). However, censoring introduces significant challenges for maximum likelihood estimation of GPD parameters. Richard L. Smith (1987) suggested that the MLE is known to be numerically unstable, particularly for the shape parameter, due to the non-concavity of the log-likelihood, data-dependent feasibility constraints, and the limited effective sample size of extreme observations .

The present thesis contributes to this literature by developing a customized maximum likelihood estimation procedure that explicitly incorporates censoring into the likelihood function. Simulation results demonstrate robustness across varying censoring thresholds and sample sizes. This contribution aligns with recent efforts to integrate EM-based strategies for incomplete data analysis (Dempster, Laird, and D. B. Rubin, 1977; Little and Donald B. Rubin, 2019). While EM algorithms have been widely used previously, existing approaches typically do not account explicitly for effective sample size in the presence of truncation and censoring. In contrast, we integrates the concept of effective sample size directly into an iterative estimation framework, thereby enhancing stability and interpretability of parameter estimates.

3. Methodology

This thesis contains an iterative algorithm for estimating the effective sample size that accommodates truncated, censored, and fully observed data. The procedure seeks to reduce bias and estimation error by alternately updating the mean and variance, while explicitly incorporating the uncertainty arising from truncation. Given that the augmented sample includes partially observed values, it is natural to embed several steps of the Expectation–Maximization (EM) algorithm within the estimation framework. The algorithm consists following components:

Expectation Step (E-step): Conditional on the current parameter estimates and the observed data, evaluate the expectation of the complete-data log-likelihood, with the expectation taken with respect to the distribution of the variables.

Maximization Step (M-step): In the M-step, the expected complete-data log-likelihood derived in the E-step is maximized with respect to the model parameters. Each iteration ensures a non-decreasing sequence of likelihood values, and convergence is attained once the increment in likelihood falls below a prespecified tolerance level ε .

Mixed log-likelihood. The mixed log-likelihood function is given by

$$\begin{aligned} \ell(\mu, \sigma^2) = & \sum_{i \in \mathcal{O}} \left[-\frac{1}{2} \log(2\pi\sigma^2) - \frac{(x_i - \mu)^2}{2\sigma^2} \right] \\ & + \sum_{j \in \mathcal{T}} \log \left[\Phi\left(\frac{U_j - \mu}{\sigma}\right) - \Phi\left(\frac{L_j - \mu}{\sigma}\right) \right] \\ & + \sum_{k \in \mathcal{C}} \log \left[1 - \Phi\left(\frac{C_k - \mu}{\sigma}\right) \right], \end{aligned}$$

Here, $\mathcal{O}$ denotes the set of fully observed observations, $\mathcal{T}$ denotes the set of truncated observations with lower and upper bounds (L_j, U_j) , and $\mathcal{C}$ denotes the set

of right-censored observations with censoring points C_k .

Fully observed data in $\mathcal{O}$ contribute the standard Gaussian log-likelihood. For truncated observations in $\mathcal{T}$, the likelihood contribution is given by the probability mass of the normal distribution over the truncation interval (L_j, U_j) . For right-censored observations in $\mathcal{C}$, the likelihood contribution corresponds to the survival probability beyond the censoring threshold C_k . The mixed log-likelihood combines these components into a unified framework.

4. Stimulated Parameters

- σ : The minimized variance, optimized with respect to the model parameters to achieve better performance in the simulated results.
- n : Number of data samples in each simulated group.
- t : The value for each observation, serving as the center for constructing its truncation or censoring interval.
- a : Lower-bound offset used to define each observation's interval
- b : Upper-bound offset used to define each observation's interval
- p : Percentage of censored observations within each group.

For each observation, the truncation or censoring interval is constructed as

$$L_i = t_i - a, \quad U_i = t_i + b.$$

which means the boundary of truncated data is not fixed.

5. Timeline

- Oct 1: finish proposal
- Dec 18: implementation complete
- Jun 9: finish final proposal
- Jul 1: first complete draft
- Jul 8: second draft
- Jul 15: last day to submit final draft to committee/second reader
- Aug 12: presentation/defense

Bibliography

- Bickel, Peter J. and Kjell A. Doksum (2001). *Mathematical Statistics: Basic Ideas and Selected Topics*. Prentice Hall.
- Chao, Anne (1987). “Estimating the population size for capture–recapture data”. In: *Journal of the American Statistical Association*.
- Coles, Stuart (2001). *An Introduction to Statistical Modeling of Extreme Values*. Springer.
- Davison, A. C. and R. L. Smith (1990). “Models for exceedances over high thresholds”. In: *Journal of the Royal Statistical Society: Series B* 52.3, pp. 393–442.
- Dempster, A. P., N. M. Laird, and D. B. Rubin (1977). “Maximum likelihood from incomplete data via the EM algorithm”. In: *Journal of the Royal Statistical Society: Series B*.
- Efron, Bradley and Vahe Petrosian (1999). “Nonparametric methods for doubly truncated data”. In: *Journal of the American Statistical Association*.
- Embrechts, Paul, Claudia Klüppelberg, and Thomas Mikosch (1997). *Modelling Extremal Events*. Springer.
- Fisher, R. A. (1941). “The negative binomial distribution”. In: *Annals of Eugenics*.
- Jennison, Christopher and Bruce W. Turnbull (2000). *Group Sequential Methods with Applications to Clinical Trials*. Chapman & Hall.
- Lagakos, S. W. (1979). “General right censoring and its impact on the analysis of survival data”. In: *Biometrics*.

- Little, Roderick J. A. and Donald B. Rubin (2019). *Statistical Analysis with Missing Data*. Wiley.
- Munasinghe, Minoli Randika (2023). “Robust Regression Models with Missing and Censored Data via the EM Algorithm”. PhD thesis. University of Thompson-Rivers University.
- Pickands, James (1975). “Statistical inference using extreme order statistics”. In: *The Annals of Statistics* 3.1, pp. 119–131.
- Sanathanan, L. (1972). “Estimating the size of a multinomial population”. In: *Annals of Mathematical Statistics*.
- Shen, Yu and Tianxi Cai (2003). “Designs for Clinical Trials with Censoring”. In: *Lifetime Data Analysis* 9.4, pp. 395–410. DOI: 10.1023/A:1025830010473.
- Smith, Richard L. (1987). “Estimating tails of probability distributions”. In: *Annals of Statistics*.